

AI shapes everything

As we close out 2024, we can look back at the massive strides AI-powered hardware has made, shaping the future of innovation in countless industries. From consumer electronics to healthcare devices and smart homes, AI hardware has gone beyond just being a buzzword.

When it comes to AI hardware, smartphones have been leading the charge. In 2024, almost every flagship model features AI-driven photography, taking the user experience far beyond simple point-and-shoot. Whether you're in low-light conditions, capturing fast-moving objects, or experimenting with bokeh effects, AI-powered camera systems have transformed smartphone photography into an intuitive, near-perfect experience.

One of the standout technologies in 2024 is computational photography, where AI algorithms automatically adjust settings for optimal results. For instance, the latest iPhones and Android devices use AI to identify scenes and objects, adjusting everything from exposure to white balance in real time. The result? Shots that look like they came straight out of a professional studio, even when you're holding the phone in one hand. AI-powered image processing is also elevating features like Night Mode, making photos taken in dark environments almost indistinguishable from daylight shots. Thanks to deep learning and neural networks, the AI can recognise different elements in your photo and optimise them individually, leading to sharp, vibrant, and well-balanced images.

The healthcare sector is another area where AI hardware is making waves. In 2024, AI-powered devices have become crucial tools for medical professionals, offering faster and more accurate diagnostics, treatment recommendations, and personalised care. While 2024 may have started as the year of experimental health gadgets, it's now the year where we've seen the first major breakthroughs in AI-assisted medical hardware. AI hardware's role in diagnostics is clear, especially with devices that analyse medical images like X-rays, MRIs,

and CT scans. Machine learning algorithms, combined with high-resolution imaging and powerful GPUs, can identify issues like tumours, fractures, or irregularities that might go unnoticed by human eyes. Devices that use AI-powered imaging software are significantly reducing diagnostic time and providing results with higher accuracy.

Moreover, wearable devices equipped with AI sensors are helping doctors remotely monitor patients. Think of smartwatches that track heart rate, blood oxygen levels, and even glucose levels, with AI running in the background, alerting users and healthcare providers to potential health risks in real time. With constant advancements in this space, AI health gadgets have evolved into something indispensable – helping not just individuals but entire healthcare systems deliver faster and more precise care. The past few years have seen a rapid growth in smart home technology, and 2024 is no exception. With AI-powered assistants and devices, homes have gotten smarter, more intuitive, and more automated. What makes this leap forward so exciting is the seamless integration between hardware and AI, making smart homes feel more natural and less like a futuristic experiment.

In 2024, AI-driven smart home security systems have gotten even more sophisticated. These systems use facial recognition, object detection, and advanced motion sensors to distinguish between familiar faces, package deliveries, and potential intruders. And thanks to AI, these systems can notify you of specific threats in real time, without flooding your phone with alerts for every little movement. AI-enhanced smart appliances have also taken household chores to the next level. Vacuum robots now feature more efficient mapping and obstacle avoidance, and they can even empty their dustbins autonomously. Refrigerators can track expiration dates of food and make shopping lists or suggest recipes based on what you have in stock. All of this is possible because of AI hardware that's continuously evolving, making our homes more energy-efficient, secure, and convenient. 